

How to Gauge a Discrete Symmetry?

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Abstract

This note is about gauging a discrete symmetry. We will first review what gauging is in a continuous case and then try to define a counterpart for a discrete symmetry. Then we try to derive some general consequences of theory after gauging.

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Gauging is the case when we couple a QFT to a gauge field (often dynamical, but some people may also call background gauge field as gauging). Apart from its importance of studying interaction between QFTs, gauging is also a powerful tool to study the symmetry of a QFT.

1 How to Gauge a Continuous Symmetry?

First we review some basics of coupling a QFT to a gauge field. Of course, we can just do the minimal coupling and be happy with it. However, to make it more general, we should not assume we always have a conserved current or even a classical lagrangian description. Then, how can we do it?

1.1 Symmetry Characterized by Generators

When we want to pin down a symmetry in QFT, the usual tools are the conserved current and the conserved charge. But these come from Noether's theorem, which already assumes two things: the symmetry is continuous, and it's invertible (so it forms a group). Neither is really essential to being a symmetry — so current and charge can't capture everything.

The Ward identity does better. A symmetry shows up as non-trivial constraints on correlation functions, and that's exactly what the Ward identity encodes. No group structure needed, so non-invertible symmetries are fine too. The catch: a Ward identity is a constraint, not an object you can build and play with. Moreover, having a constraint on correlation function doesn't necessarily mean we have a symmetry, for example, if a symmetry has an anomaly, the correlation function indeed satisfy some constraints, but we think of it as the symmetry is broken.

So what we'd really like is an “object-like” thing — like a current or charge — that's general enough to describe any symmetry. Good news: it exists. It's the **symmetry generator**, and the trick is that it's a topological operator. That's the property that sets us free from continuity and invertibility, and it's what we'll build the definition around.

In what follows, we'll let our intuition about continuous symmetries lead the way — and watch the conserved charge turn into a topological operator.

1.1.1 Classical Definition

We first take an example of a classical field theory with a $U(1)$ symmetry (say a complex scalar field). Noether's theorem tells us that there is a conserved current j satisfying the conservation equation:

$$d * j = 0 \tag{1}$$

And we can define a conserved charge by integrating the current over a spatial slice:

$$Q = \oint_{\Sigma} * j \tag{2}$$

where Σ is a spatial slice. Now we can try to integrate the current over a codimension-1 manifold M_{d-1} , which is not necessarily a spatial slice, and define a charge as:

$$Q(M_{d-1}) = \oint_{M_{d-1}} * j \tag{3}$$

and a related classical quantity we called the **symmetry generator**:

$$U(M_{d-1}) = \exp(i\alpha Q(M_{d-1})) \tag{4}$$

One can prove that these objects are topological, which means that they are invariant under continuous deformation of the manifold M_{d-1} , as long as we don't cross any strange stuffs. This can be proven with the stokes theorem:

$$Q(M_{d-1}) = \oint_{M_{d-1}} *j = \oint_{M'_{d-1}} *j + \oint_{B_d} d*j = Q(M'_{d-1}) \quad (5)$$

where B_d is the manifold connecting M_{d-1} and M'_{d-1} satisfying:

$$\partial B_d = M_{d-1} \sqcup M'_{d-1} \quad (6)$$

1.1.2 Quantum Interpretation

Now we generalize the classical object we define above to a quantum one. Notice that the world is NOT classical, it is definitely true that we can have some quantum object without classical counterparts. Namely can't be derived from quantizing some classical object using single quantization scheme (eg. canonical quantization), and the symmetry generator is such an object.

Thus, we don't derive a quantum theory from a classical one, but use different quantization scheme and results to justify how a quantum symmetry generators exit what properties should they have.

The properties of **quantum symmetry generators** (and justifications) are:

1. They are **Topological Operators**. This can be justified by the classical conservation equation, plugging them into a path integral won't change this fact. Moreover, as a quantum theory, in the canonical picture if we want to define these symmetry generators as line operators, they have to commute with the Energy-Momentum tensor, which generates the spacetime translation.
2. If inserted in a time slice, they act as an operators on the Hilbert space in the canonical formalism. This can be justified by analoging the ordinary symmetry charge defined on a time slice.
3. If inserted in vetically in the path integral, they behave as a defect, which twisted the Hilbert Space. This can't be justified straightforwardly, but a few lines of calculation in the path integral formalism:

YL: [\[finish this part later\]](#)

(2 and 3 properties are called the operator/defect principles)

4. When going through local operators, they act as a symmetry transformation on the local operators. This can be justified by the Ward identity.

YL: [\[finish this part later\]](#)

1.1.3 A Slogan for Symmetry

The current/charge picture is tied to continuous symmetries with a classical Lagrangian. To go beyond it, we adopt the following slogan.

Definition 1.1

Symmetry Slogan

A symmetry is characterized by the existence of a topological operator — a *symmetry generator*.

This is a powerful and sensible generalization. As we will see, it applies equally to continuous symmetries, discrete symmetries, and even symmetries without an inverse, the so-called *non-invertible symmetries*.

1.2 What is a Gauge Field?

Now we have a grasp of what is a symmetry, we turn to the other question of gauging: what is a gauge field?

Naively, we define a gauge field as a Lie algebra valued one form field:

$$A = A_\mu dx^\mu \text{ where } A_\mu \in \mathfrak{g} \quad (7)$$

However, this is generally not the case. A gauge field may not be globally defined.

A classical and experimental example is the AB effect of U(1) gauge field, where we know that:

$$\exp\left(i \oint_C A dl\right) \quad (8)$$

and is exactly an experimental observable. Thus we need to rethink the definition of gauge field. The critical point is that a gauge field may not be defined globally, but only on patches, while field on different patches are related by a suitable gauge transformation.

Moreover, from demanding consistency in AB phase and a globally well defined curvature (indeed we measure electric and magnetic fields and don't see any singularity), we need to have a quantization condition on the flux of the gauge field:

$$\oint_S F \in 2\pi\mathbb{Z} \quad (9)$$

1.2.1 Mathematical Characterization

A mathematical framework, namingly **fibre bundle** is developed to characterize all these features of gauge field (which I wish to introduce in another note). Now to make things simple, we try to construct a gauge field step by steps:

- **Prepare Patches of Principle G Bundle:**

1. Given a manifold M_d choose a set of open over $\{U_i\}$
2. On each patch U_i we “choose a trivialization of the Principle G-bundle”. This is kinds of equivalent to another fancy name, which is choosing a **section** of the bundle, which is a map from the base space to the total space of the bundle:

$$s_i : U_i \rightarrow P \quad (10)$$

well we can just understand as mapping each point in the patch to a group element and its only defined locally. Physically this is equivalent to choosing a gauge, as discussed before, we can't choose a global gauge, thus we can only choose a local gauge on each patch.

3. On the overlap of two patches U_i and U_j , we have a transition function:

$$g_{ij} : U_i \cap U_j \rightarrow G \quad (11)$$

this relates the two trivialization on the two patches, and we have:

$$s_i = g_{ij} s_j \quad (12)$$

4. On a triple overlap of three patches U_i , U_j and U_k , we have a consistency condition:

$$g_{ij} g_{jk} g_{ki} = 1 \quad (13)$$

- **Construct Gauge Field:**

gauge field can be understood as a connection on the principle G-bundle after a pullback by the section we choose. But we can just understand it as a well defined Lie algebra valued one form on each patch:

$$A^{(i)} \in \Omega^1(U_i) \otimes \mathfrak{g} \quad (14)$$

and on the overlap of two patches, we have a gauge transformation:

$$A^{(j)} = g_{ij}^{-1} A^{(i)} g_{ij} + g_{ij}^{-1} dg_{ij} \quad (15)$$

Two important objects of the principle bundle are the curvature and the holonomy of the gauge field, which are defined as:

$$F^{(i)} = dA^{(i)} + A^{(i)} \wedge A^{(i)} \quad (16)$$

$$\text{Hol}(C) = \exp \left(i \oint_C A \right) \quad (17)$$

physically, they correspond to the field strength and the AB phase of U(1) gauge field.

1.2.2 Quantization of Flux

With suitable consistent mathematical construction of gauge field from fibre bundles, we can see that quantized flux arises naturally from mathematical properties of fibre bundles.

- Quantization of U(1) flux is related to the first Chern class of the U(1) bundle, which is an integer.

$$c_1 = \frac{1}{2\pi} \oint_S F \in \mathbb{Z} \quad (18)$$

- Quantization of SU(N) non-abelian flux is related to the second Chern class of the non-abelian bundle, which is also an integer.

$$c_2 = \frac{1}{8\pi^2} \oint_S \text{Tr}(F \wedge F) \in \mathbb{Z} \quad (19)$$

1.2.3 Higher Form Gauge Field

This note we try to touch some generalization of symmetry to higher form. Thus, we need to define what is a higher form gauge field. A theorem shows that:

Theorem 1.2

Abelian Higher Form Symmetry

A higher form symmetry is always an abelian symmetry, thus the gauge field of a higher form symmetry is always an abelian gauge field.

To make life simple, we here just consider **U(1) higher form symmetry** and **U(1) higher form gauge field**, which is a common case.

For a p-form symmetry, the current is a (p+1)-form j , thus it should couple to a (p+1)-form U(1) gauge field A as:

$$S = \int_{M_d} A \wedge * j \quad (20)$$

where A is a (p+1)-form gauge field, and the gauge transformation is:

$$A \rightarrow A + d\Lambda_p \quad (21)$$

And here Λ_p is a **p-form Gauge Field** : Remember the gauge transformation is a gauge field!!!

For A as a $(p+1)$ -form gauge field, the field strength is a $(p+2)$ -form, that are quantized under the following condition:

$$F = dA, \quad \oint_{S_{p+2}} F \in 2\pi\mathbb{Z} \quad (22)$$

it is quantized for any $p+2$ dimensional closed manifold S_{p+2} . Moreover, we should notice that the gauge transformation is also a gauge field. Thus there is a “gauge transformation of gauge transformation”, which is a $(p-1)$ -form gauge field Λ_{p-1} :

$$\Lambda_p \rightarrow \Lambda_p + d\Lambda_{p-1} \quad (23)$$

and the gauge transformation have a quantized flux condition as well:

$$\oint_{S_{p+1}} d\Lambda_p \in 2\pi\mathbb{Z} \quad (24)$$

This is nothing strange to us, consider the ordinary $U(1)$ gauge field, the gauge transformation is:

$$A \rightarrow A + d\Lambda_0 \quad (25)$$

and if we integrate the gauge transformation over a closed loop, we have:

$$\oint_{S_1} d\Lambda_0 \in 2\pi\mathbb{Z} \quad (26)$$

This is due the fact if we integrate through patches, we can have a $2\pi\mathbb{Z}$ shift, by the above construction.

YL: [In fact I really don't think I understand this, this should be taught in a cohomology language naturally, which I don't understand.]

1.3 How to Couple to a Gauge Field?

Now we start with the true topic, Gauging. By gauging a symmetry, we mean coupling the theory to a gauge field and make it dynamical. This section is to point out an observation. Namely, coupling to a gauge field is equivalent to inserting the symmetry generator properly.

1.3.1 BG Gauge Field Insertion of Symmetry Generator

1.3.2 AB phase as going through Symmetry Generator

Now we consider local operators in the theory. Notice that the local charged operators are no longer gauge invariant observables but rather operators with Wilson line attached.

1.4 Anomaly

For a system with a lagrangian description, we may expect a symmetry of the lagrangian may not be a symmetry of the QFT, say, the corresponding Ward identity is not satisfied. This is what we call an **anomaly**.

1.4.1 Definition and Classification

Due to the discussion of symmetry and background gauge field, we can give a definition from the perspective of gauging.

1.4.2 Anomaly inflow and Invertible Theory

2 How to Gauge a Discrete Symmetry?

Now we turn to the main topic of this note, gauging a discrete symmetry. To do this we may first define what we mean by a discrete gauge field, and a discrete symmetry of a QFT.

2.1 Discrete Symmetry

2.1.1 Traditional Characterization

Wigner Theorem tells us that a Quantum theory has a symmetry is given by a Unitary or Anti-Unitary operator, forming a representation of the symmetry group on the Hilbert space. We can characterize the symmetry with these operators.

A discrete symmetry is a symmetry whose symmetry group is a discrete group, which can be finite or infinite. The Hilbert space forms a Unitary or Anti-Unitary representation of the symmetry group, and those operator

- **Group Structure:**

$$U(g_1)U(g_2) = U(g_1g_2) \quad (27)$$

- **Conserved:**

$$U(g)HU(g)^{-1} = H \quad (28)$$

- **Generate transformation:**

$$U(g)O(x)U(g)^{-1} = R(g)O(x) \quad (29)$$

- **Ward Identity:** they also have some Ward identities, which can be used to constrain the correlation function. A simple example is the $\mathbb{Z}_2$ symmetry of free scalar field theory, which acts as $\varphi(x) \rightarrow -\varphi(x)$, and the Ward identity tells us that the correlation function of odd number of φ must vanish.
- ...

2.1.2 Modern Characterization

Remember the slogan we introduced before, we can generalize to characterize symmetries with symmetry generators. Imposing consistency conditions and observing the relation with Ward identities, we can find out all these symmetry generators and identify them with symmetries.

As far as I know, in 2D QFT these generalized symmetries can be described by Topological Defect Lines systematically [1], satisfying a series of characteristic properties. It turns out that these symmetries doesn't have a group structure, but rather a structure of fusion category. For higher dimensional QFT, the story is more complicated and general.

2.2 Discrete Gauge Field

Another problem is what is a discrete gauge field and what mathematical language characterize these objects. For discrete symmetry we don't have a Lie algebra, the Lie algebra of a discrete group is 0 thus:

$$A = 0, \quad F = 0 \quad (30)$$

Naively we only have trivial objects that are meaningless. However, we remember that a gauge field may not be globally defined, but only on patches. Though we have 0 for the gauge field locally, we may have non-trivial holonomy when going through patches:

$$\text{Hol}(C) = \exp\left(i \oint_C A\right) = g \in G \quad (31)$$

In fact to characterize this data, we don't need to introduce the gauge field but the transition function between patches:

$$g_{ij} : U_i \cap U_j \rightarrow G \quad (32)$$

3 Consequence Under Gauging

Bibliography

- [1] C.-M. Chang, Y.-H. Lin, S.-H. Shao, Y. Wang, and X. Yin, “Topological Defect Lines and Renormalization Group Flows in Two Dimensions,” *Journal of High Energy Physics*, vol. 2019, no. 1, p. 26, Jan. 2019, doi: [10.1007/JHEP01\(2019\)026](https://doi.org/10.1007/JHEP01(2019)026).